# Data Flow Diagram

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|---|---|
| **Date** | 29 September 2026 |
| **Team ID** | SWTID-2026-8642 |
| **Project Name** | Import Data Using Transform Maps (Spreadsheet) |
| **Maximum Marks** | 2 Marks |

## Data Flow Diagram

Create a Data Flow Diagram (DFD) to show how data moves through your system - between external entities, processes, and data stores. Use the symbol legend below as a guide.

## DFD Symbol Legend

<table>
<tr><th>Symbol</th><th>Name</th><th>Description</th></tr>
<tr><td>Oval / Rounded shape</td><td>External Entity</td><td>
<ul>
<li>Data Owner / Analyst: Uploads the raw spreadsheet file and initiates the import.</li>
<li>System Administrator: Configures transform maps, field mappings, and coalesce rules.</li>
<li>Target Application System: Receives and stores the fully processed records.</li>
</ul>
</td></tr>
<tr><td>Rectangle with numbered header</td><td>Process</td><td>
<table>
<tr><th>Process ID</th><th>Process Name</th><th>Description</th></tr>
<tr><td>1.0</td><td>Upload & Parse Spreadsheet</td><td>Reads the raw Excel/CSV file and parses rows</td></tr>
</table>
</td></tr>
</table>

<table>
<tr><td></td><td></td><td>
<table>
<tr><td></td><td></td><td>into staging table format.</td></tr>
<tr><td>2.0</td><td>Map & Transform Data</td><td>Maps source columns to target fields using script rules and coalesce parameters.</td></tr>
<tr><td>3.0</td><td>Load Data & Log Errors</td><td>Inserts valid records into the target database and writes execution errors to logs.</td></tr>
</table>
</td></tr>
<tr><td>Rectangle (solid fill, no number)</td><td>Data Store</td><td>
<table>
<tr><th>Data Store ID</th><th>Data Store Name</th><th>Description</th></tr>
<tr><td>D1</td><td>Import Set Staging Table</td><td>Holds raw, unprocessed spreadsheet data temporarily.</td></tr>
<tr><td>D2</td><td>Transform Map Config</td><td>Stores column mapping definitions, transformation scripts, and key rules.</td></tr>
<tr><td>D3</td><td>Target Database Table</td><td>The final repository where updated and new records reside.</td></tr>
<tr><td>D4</td><td>Import Set Logs</td><td>Stores execution history, error records, and</td></tr>
</table>
</td></tr>
</table>

| | | |
|---|---|---|
| | | skipped row details. |
| **Labeled arrow** | **Data Flow** | **Source** / **Process / Destination** / **Data Passed (Data Flow Label)**<br>**Data Owner** / **Process 1.0** / Raw Spreadsheet File (Excel / CSV)<br>**Process 1.0** / **Data Store D1** / Staged Import Set Records<br>**System Admin** / **Data Store D2** / Transform Rules & Field Mappings<br>**Data Store D1 & D2** / **Process 2.0** / Staged Records & Mapping Configurations<br>**Process 2.0** / **Process 3.0** / Transformed & Validated Records<br>**Process 3.0** / **Data Store D3** / Final Target Records (Insert / Update)<br>**Process 3.0** / **Data Store D4** / Error Logs & Execution Summaries<br>**Data Store D4** / **Data Owner** / Status Reports & Error Alerts |

**Example**

The diagram below illustrates how the symbols above come together to form a complete data flow diagram.

